

Module 1: Work Order Management - Complete Specification

Tab 3: Site Survey

Version: 2.0
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1. Purpose

This tab is designed for surveyors to capture detailed on-site information for excavation work. It combines pre-populated data from the Work Order with interactive tools for surveyors to input measurements, capture GPS coordinates, create a visual representation of the planned work, and document the site with photos. This form serves as the primary field survey record before excavation permits are obtained.

2. Section 1: General Information (Read-Only)

This section displays key information autopopulated from the **Work Order Information** tab to provide context for the surveyor. These fields are not editable in this tab.

Field Name (English)	Data Source	Field Name (Arabic)
Work Order No.	<u>work_orders.work_order_number</u>	رقم أمر العمل
District	<u>work_orders.district</u>	الحي
Station Number	<u>work_orders.station_number</u>	رقم المحطة
Latitude	<u>work_orders.latitude</u>	خط العرض
Longitude	<u>work_orders.longitude</u>	خط الطول

3. Section 2: Survey Details (Input)

This section contains fields that the surveyor must complete.

Field Name (English)	Data Type	Notes	Field Name (Arabic)
Surveyed Station Number	Integer	For verification purposes.	رقم المحطة التي تم مسحها
Survey Date	Date	Defaults to the current date.	تاريخ المسح
Permit Type	Checkbox Group	Options: Normal Excavation, Single Connection, Multiple Connection.	نوع التصريح

4. Section 3: Drawing Area & Excavation Details

This section combines the interactive map with the details table, as they are directly linked.

4.1. Drawing Area (Interactive Map)

The Drawing Area is an interactive map-like interface where surveyors visually map out the excavation work. It includes the following controls:

- **Capture Point Button**: When clicked, this records the device's current GPS coordinates and places a point on the map. If it's the second point for a given line, it draws a straight line connecting it to the previous point and automatically calculates the length, which is populated in the corresponding row of the Excavation Details table.
- **Add Line Button**: Adds a new row to the Excavation Details table below, allowing the user to start capturing points for a new line segment.
- **Map Layer Toggle**: A button to switch a satellite map background layer on or off.
- **Icon Toolbar**: A toolbar with a selection of icons representing electrical equipment (e.g., RMU, Transformer, Unit, Pole). The surveyor can select an icon and click on the map to place it. Placed icons and lines should be selectable and movable.
- **Point Interaction**: When a captured point on the map is clicked, a pop-up or side panel appears with the following options:
 - **Capture Photo Button**: Opens the device's camera. Up to 4 photos can be captured for each point.
 - **Photo Gallery**: Displays thumbnails of the photos taken for that point.

4.2. Photo Capture Requirements

- **Metadata Overlay**: Every photo captured through the app must have the following information automatically embedded as a watermark at the bottom right of the image:
 - Date (YYYY-MM-DD)
 - Time (HH:MM:SS)
 - GPS Coordinates (Lat, Long)
 - A compass icon or indicator showing the cardinal direction the photo was taken.
- **File Naming Convention**: Photos must be saved with a standardized file name: [Request Number]-[Work Type]-[Point#]-[Photo#].jpg. For example: 241053081-804-Point2-1.jpg.

4.3. Excavation Details Table

This table lists the line segments drawn on the map. Rows are added by clicking the Add Line button in the Drawing Area.

Column Header	Data Type / Control	Functionality	Field Name (Arabic)
Line	Text (Read-only)	Automatically numbered (A1, A2, A3, ...).	المسار
Length (m)	Number (Read-only)	Automatically calculated and populated after two points for a line are captured in the Drawing Area.	الطول
Depth (m)	Number (Input)	Surveyor manually enters the excavation depth.	العمق
Width (m)	Number (Input)	Surveyor manually enters the excavation width.	العرض
Milling Type	Dropdown	A predefined list of milling types.	نوع الحفر
Road Type	Dropdown	A predefined list of road surface types (e.g., Asphalt, Tiles, Soil).	نوع المسار

5. Section 4: Approvals & Signatures

This section is for final verification and sign-off of the site survey.

Work Type Checkboxes: A series of checkboxes for the surveyor to indicate the nature of the work:

- ☐ Install New MP Board (تركيب لوحة توزيع جديدة)
- ☐ Existing MP Board (لوحة توزيع قائمة)
- ☐ Install New TF (تركيب محول جديد)
- ☐ Connection From Pole (توصيل من عمود)
- ☐ New Road & Developed (شارع جديد ومطور)
- ☐ Old Road & Undeveloped (شارع قديم وغير مطور)

Signature Pads: Two digital signature pads for electronic sign-off:

- 1 **Consultant Representative:** Name Field (اسم مندوب الاستشاري) and Signature Area (التوقيع) - a canvas for drawing with a mouse or stylus.
- 2 **Contractor Surveyor:** Name Field (اسم مساح المقاول) and Signature Area (التوقيع) - a canvas for drawing with a mouse or stylus.

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